

Atharv Agashe

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EDUCATION

Dartmouth College, Hanover, NH

June 2025

B.A. Computer Science and Mathematics (Double Major), GPA: 3.72 *cum laude*

EXPERIENCE

DraftKings

Software Engineer

July 2025 – Present

- Re-architected the Marketing Content Registry as an SDK and **built an agentic layer** for editing, translating, auditing, and validating templates; **onboarded 20+ developers** and supported **\$180 million** in promotions.
- Automated promotion localization by building an agent translation harness that **processed 95k promotions**, **reduced required stakeholders from 45+ to 2**, and cut template revisions from 15+ to a median of 1. (**OpenAI API**)
- Scaled the email notification service from **20 to 1k RPS** while **reducing p99 latency from 200 ms to 34 ms**; served **243 million requests** with **'8 nines' of successful responses**. (**k6, APM**)
- Eliminated a **0.5 GB/day** memory leak that caused weekly pod restarts, enabling horizontal scaling while **reducing the compute fleet by 10x**. (**gcdump**)
- Instrumented AI usage across the Marketing Content SDK and developed 5 Claude skills targeting the largest workflow bottlenecks, **reducing template development time by 80%+** (**Datadog**)
- Built personalized content generation and end-to-end attribution infrastructure projected to **support \$800 million** in promotions, creating a production feedback dataset for evaluating and training future RL models
- Led team AI initiatives by automating deployments, smoke tests, and service-health monitoring; eliminated production incidents and reduced release-related bug resolution time by 40%+ using Datadog traces and code diffs (via MCP)

Software Engineering Intern

June 2024 – August 2024

- Increased annual revenue by **\$35 million** by building a standard internal promotion structure across all DK products to enable precise user segment targeting, enabling cross-selling, and optimizing promo conversions (**.NET, C#, SQL**).
- Raised team velocity by 30%** by refactoring the database migration architecture to automate schema changes. Built code-gen tools to **reduce schema related development time by 95%**, while increasing reliability (**SQL, AWS**).

Amazon (AWS)

May 2023 – August 2023

Software Engineering Intern

- Identified and consolidated the **top 0.004%** (~1 GB) of critical metrics from a **500 PB S3 bucket** to help access trends for code base health.
- Improved database read speeds by 400%+** by optimizing database design with indexing techniques. Saved 5 person engineering team **4 hrs/week of post-mortem analysis time** (**SQL, Hibernate**).

SmartMicroscope (DALI Lab)

January 2023 – June 2025

Software Lead (8 person research team)

- Developed a smart microscope that enabled pathologists to integrate ML-processed slide samples in their regular workflow. **Saved 3 days per diagnosis** on average, a **65% reduction** compared to traditional microscopes

PUBLICATIONS

ACM (Association of Computing Machinery): Volume 44, Issue 6 of TOG on Acousto-Optic Beam Steering

PROJECTS

Arbitrage and MM Bot: Generated **>\$380k in profit** to date. Since 2020, built a variety of decentralized finance (DeFi) models, arbitrage bots, and automated delta neutral MM strategies that turned a **principle of 25k into >\$400k**.

AI Chess Trainer: I **gained 350 ELO (from 89.8% percentile to 99.1% percentile)** by building a Discord bot that scrapes a player's recent chess games to identify trends in gameplay and sends daily puzzles targeting player weaknesses.

SKILLS

Technical Skills: Frontend (HTML, CSS, JS), Backend (.NET, Python, Java, C, SQL, S3, AuroraDB), Libraries/Tools (AWS Bedrock, LlamaIndex, OpenAI API, PyTorch, Scikit-Learn, ReactJS, TailwindCSS, Selenium, Git, OpenCV, Pandas)

Hobbies: Chess (ELO 2120), Poker, Photography, Squash, Rock Climbing, Karate (Black Belt), Guitar